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Matrix-based vs. vector-based linear discriminant analysis: A comparison of regularized variants on multivariate time series data

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ABSTRACT

Over the past two decades, matrix-based or bilinear discriminant analysis (BLDA) methods have received much attention. However, it has been reported that the traditional vector-based regularized LDA (RLDA) is still quite competitive and could outperform BLDA on some benchmark datasets. A central question is whether the superiority of the vector-based RLDA would always hold for general matrix data, or is there any type of matrix data on which BLDA would perform better than RLDA? Actually, the reported comparisons are found to suffer from two limitations: (i) the comparisons are only limited to image data, and (ii) regularized RLDA is compared with non-regularized BLDA. In this paper, we break the two limitations and investigate the central question on another type of matrix data, namely multivariate time series (MTS) data. We propose a new two-parameter regularized BLDA (RBLDA) for MTS data classification. To choose the two parameters, we develop an efficient model selection algorithm. The newly proposed RBLDA enables us to perform a fair comparison between vector-based RLDA and matrix-based RBLDA. Experiments on a number of real MTS data sets are conducted to compare RBLDA with RLDA and evaluate the proposed algorithm. The results reveal that the superiority of the vector-based RLDA does not always hold for general matrix data, and RBLDA outperforms RLDA on MTS data. Moreover, the proposed model selection algorithm is efficient, and RBLDA can produce better visualization of MTS data than RLDA.

1. Introduction

Principal component analysis (PCA) and Fisher linear discriminant analysis (LDA) are widely used dimension reduction methods for vector data, where observations are vectors. To perform dimension reduction on matrix data, where observations are matrices, one common solution is first to vectorize the matrix data and then applying PCA and LDA to the resulting data. However, the vectorization has some disadvantages. First, it loses the natural matrix data structure [1]. Second, the vector-based methods may suffer from the high-dimensional problem [2].

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To deal with the two disadvantages of image data, instead of employing vectorization, several matrix-based dimension reduction methods have been proposed, such as bilinear PCA (BPCA) [3], bilinear discriminant analysis (BLDA) [4,5], etc. The common feature is that these methods use bilinear transformations, rather than the linear ones in PCA and LDA. As a result, their numbers of parameters are much less than those in PCA and LDA and can greatly alleviate the high-dimensional problem and significantly reduce computational costs [1]. Despite the advantages of matrix-based methods, when would matrix-based methods be superior to vector-based counterparts? When would vector-based methods be better? These are important issues of long-standing interest. In essence, the matrix-based BPCA and BLDA [4,5] can be viewed as a special case of the vector-based PCA and LDA under the assumption that the transformation matrix is separable [2,5]. Naturally, when this assumption tends to hold approximately, it would be expected that matrix-based methods can perform favorably in finite matrix data applications since they simply require specifying a much smaller number of parameters.

Many efforts have been made to address the important issues. The earlier general view is that matrix-based methods could be better than vector-based cousins in small sample size cases. Thus a key factor to the superiority is the small-sample-size training. This view holds for PCA-related methods, even on datasets where the assumption does not hold, the matrix-based BPCA could obtain better performance than the vector-based PCA due to the variance-bias trade-off, e.g., [2]. However, this view may not be applicable to LDA-related methods, because the vector-based regularized LDA (RLDA) outperforms the matrix-based BLDA on a number of real image datasets, as shown by the comprehensive study in [6]. Similar results can be found in [7].

A central question is whether the superiority of RLDA over BLDA would always hold for general matrix data, or is there any type of matrix data on which the assumption of a separable transformation matrix could be more reasonable and hence BLDA would perform better than RLDA? In fact, it is found that the reported empirical comparisons between BLDA and RLDA in [6,7] have two limitations: (i) the comparisons are only limited to image data; and (ii) regularized RLDA is compared with non-regularized BLDA.

In this paper, we focus on another type of matrix data, namely multivariate time series (MTS) data, whose observation is usually the result of multiple variables measured at different time points. MTS data classification is a hot research topic in time series data mining. Many methods have been proposed in the literature, including discriminant analysis [8], hypothesis testing [9], singular value decomposition (SVD) [10], common PCA [11], dynamic time warping (DTW) [12,13], Mahalanobis distance-based DTW [14], etc. However, all of these methods fail to consider the matrix data structure inherent in MTS data. Images and MTS are two different types of matrix data, because the both modes (i.e., row and column) of images are variables (i.e., pixels), while one mode of MTS data represents variable and the other is time. Intuitionally, it seems that the assumption of a separable transformation matrix in BPCA and BLDA is more suitable for such data as the two modes of MTS are different. In fact, BPCA has been suggested for MTS data in [15]. Since BPCA is an unsupervised method, BLDA and its extension based on pseudo-inverse (PBLDA) are further suggested in [16] to utilize the matrix data structure and label information simultaneously. Their empirical results show that these matrix-based methods are often advantageous over closely related methods. However, to our knowledge, the interesting comparison with the well-known RLDA is still missing.

To break the second limitation, we need to develop a new regularized BLDA (RBLDA) so that we can perform a fair empirical comparison with RLDA. Note that we have used a variant called one-parameter RBLDA in [7]. However, the research focus in [7] is very different from that in this paper. First, only image data is considered and the objective is to improve the performance of vector-based RLDA on image data by using the one-parameter RBLDA as an immediate step for initial dimension reduction. Second, the regularization parameter in the one-parameter RBLDA is simply set to be a fixed value and how to choose it efficiently is not considered.

Inspired by the encouraging results of matrix-based methods on MTS data, in this paper, we break the two limitations and investigate the central question. The main contributions of this paper are threefold as follows.

(i) We propose a new two-parameter RBLDA for MTS data classification. Furthermore, to choose the two parameters with cross validation, we develop an efficient model selection algorithm so that the computational burden suffered by the conventional cross validation procedure can be alleviated substantially, especially in the case that the number of variables or time points is much greater than the other. This breaks the second limitation.
(ii) With our newly developed two-parameter RBLDA, the only factor that affects the performance is mainly the data type, namely MTS data. This enables us to conduct a fair empirical comparison between vector-based RLDA and matrix-based RBLDA on MTS data. This breaks the first limitation. The results show that the superiority of RLDA over BLDA does not always hold for general matrix data and RBLDA generally performs better than RLDA on MTS data.
(iii) We further explore the visualization of MTS data using RBLDA. Such visualization is rarely considered with existing matrix-based methods.

The remainder of the paper is organized as follows. Sec. 2 gives a brief review of linear discriminant analysis (LDA), regularized LDA (RLDA), and the existing implementation of BLDA. Sec. 3 proposes our RBLDA and its efficient model selection algorithm. Sec. 4 constructs an empirical study to evaluate the proposed algorithm and compare RBLDA with several closely related competitors. We end the paper with some conclusions and discussions in Sec. 5.

Notations: in the sequel, the transpose of vector/matrix is denoted by the superscript $'$, and the $d \times d$ identity matrix by $\mathbf{I}_d$. Moreover, $\mathrm{tr}(\mathbf{A})$ is the trace of matrix $\mathbf{A}$ and $\mathbf{B} = \mathrm{blkdiag}(\mathbf{A}_1, \ldots, \mathbf{A}_m)$ stands for the block diagonal matrix formed by aligning the input matrices $\mathbf{A}_1, \ldots, \mathbf{A}_m$ along the diagonal of $\mathbf{B}$.

2. Review of LDA and related variants

2.1. Linear discriminant analysis (LDA)

Given a set of d-dimensional vector-valued data $\{\mathbf{x}_i\}_{i=1}^n$ consisting of c classes, let $\mathbf{m}_k = \frac{1}{n_k}\sum_{i\in C_k}\mathbf{x}_i$ be the sample mean of class k, C_k, and $\mathbf{m} = \frac{1}{n}\sum_{i=1}^n \mathbf{x}_i$ be the sample mean, then the between-class, within-class and total scatter matrices are given by

$$\mathbf{S}_b = \frac{1}{n}\sum_{k=1}^c n_k(\mathbf{m}_k - \mathbf{m})(\mathbf{m}_k - \mathbf{m})', \tag{1}$$

$$\mathbf{S}_w = \frac{1}{n}\sum_{k=1}^c \sum_{i\in C_k}(\mathbf{x}_i - \mathbf{m}_k)(\mathbf{x}_i - \mathbf{m}_k)',$$

$$\mathbf{S}_t = \frac{1}{n}\sum_{i=1}^n (\mathbf{x}_i - \mathbf{m})(\mathbf{x}_i - \mathbf{m})'. \tag{2}$$

Consider a linear transformation $\mathbf{y} = \mathbf{V}_w'\mathbf{x}$, where $\mathbf{V}_w \in \mathbb{R}^{d\times q}$ and $q < d$. The within-class and between-class scatter matrices in $\mathbf{y}$-space are $\mathbf{V}_w'\mathbf{S}_w\mathbf{V}_w$ and $\mathbf{V}_w'\mathbf{S}_b\mathbf{V}_w$, respectively. The Fisher criterion aims to find $\mathbf{V}_w$ that maximizes $\mathcal{F}$ as [17],

$$\mathcal{F} = \max_{\mathbf{V}_w} \operatorname{tr}\left\{(\mathbf{V}_w'\mathbf{S}_w\mathbf{V}_w)^{-1}(\mathbf{V}_w'\mathbf{S}_b\mathbf{V}_w)\right\}.$$

It can be seen that $\mathbf{V}_w$ can only be determined up to a $q\times q$ nonsingular transformation. A common solution is obtained by solving the problem

$$\max_{\mathbf{V}_w} \operatorname{tr}\left\{\mathbf{V}_w'\mathbf{S}_b\mathbf{V}_w\right\}, \quad s.t.\ \mathbf{V}_w'\mathbf{S}_w\mathbf{V}_w = \mathbf{I}_q, \tag{3}$$

and the columns of $\mathbf{V}_w$ are $\mathbf{S}_w$-orthogonal due to the constraint $\mathbf{V}_w'\mathbf{S}_w\mathbf{V}_w = \mathbf{I}_q$.

Alternatively, $\mathbf{S}_w$ in (3) can be replaced by $\mathbf{S}_t$ [18,19] and the optimization problem is

$$\max_{\mathbf{V}_t} \operatorname{tr}\{\mathbf{V}_t'\mathbf{S}_b\mathbf{V}_t\}, \quad s.t.\ \mathbf{V}_t'\mathbf{S}_t\mathbf{V}_t = \mathbf{I}_q, \tag{4}$$

and then the columns of $\mathbf{V}_t$ are $\mathbf{S}_t$-orthogonal due to the constraint $\mathbf{V}_t'\mathbf{S}_t\mathbf{V}_t = \mathbf{I}_q$. Note that the above two problems have close relationship as detailed in Proposition 1, e.g., $\mathbf{V}_w$ can be obtained via $\mathbf{V}_t$ by (9).

2.2. Regularized LDA (RLDA)

For high dimensional data, where the data dimension d is greater than the sample size n, $\mathbf{S}_t$ or $\mathbf{S}_w$ is singular and hence LDA can not be performed. In addition, for the data whose variables are highly correlated even if d is less than n, $\mathbf{S}_t$ or $\mathbf{S}_w$ could be approximately singular and the performance of LDA could be degenerated greatly. The regularized LDA (RLDA) proposed in [20] is a popular method to tackle these two problems simultaneously. The corresponding optimization problem to (4) in RLDA is

$$\max_{\mathbf{V}_t} \operatorname{tr}\{\mathbf{V}_t'\mathbf{S}_b\mathbf{V}_t\}, \quad s.t.\ \mathbf{V}_t'\mathbf{S}_t^r\mathbf{V}_t = \mathbf{I}_q, \tag{5}$$

where $\mathbf{S}_t^r = (1-r)\mathbf{S}_t + r\hat{\sigma}^2\mathbf{I}_q$, $\hat{\sigma}^2 = \operatorname{tr}(\mathbf{S}_t)/d$ and the regularization parameter $r \in (0,1]$. Clearly, $\mathbf{S}_t^r$ is shrunk to a scalar covariance. The parameter r is usually determined via cross validation.

By the method of Lagrange Multipliers, the closed form solution $\mathbf{V}_t$ can be obtained by solving the generalized eigenvalue problem

$$\mathbf{S}_t^{r-1}\mathbf{S}_b\mathbf{V}_t = \mathbf{V}_t\mathbf{\Lambda}_t, \tag{6}$$

under the constraint $\mathbf{V}_t'\mathbf{S}_t^r\mathbf{V}_t = \mathbf{I}_q$.

Next we analyze the main time complexity of RLDA implemented by (6). The formation of $\mathbf{S}_t^r$ costs $O(d^2n)$ and its inverse takes $O(d^3)$. The eigenproblem (6) also costs $O(d^3)$. Thus the total cost of RLDA implemented by (6) is $O(d^2n) + O(d^3)$. When $d \gg n$, this cost could be very heavy. In Sec. 2.4, we review a computationally efficient implementation of RLDA.

2.3. The relationship between the two problems formulated by $\mathbf{S}_w$ and $\mathbf{S}_t$

In this subsection, we give the relationship between the two problems formulated by $\mathbf{S}_w$ and $\mathbf{S}_t$, which seems missing in the literature.

The corresponding optimization problem to (3)

$$\max_{\mathbf{V}_w} \operatorname{tr}\{\mathbf{V}_w'\mathbf{S}_b\mathbf{V}_w\}, \quad s.t.\ \mathbf{V}_w'\mathbf{S}_w^r\mathbf{V}_w = \mathbf{I}_q, \tag{7}$$

where $\mathbf{S}_w^r = (1-r)\mathbf{S}_w + r\hat{\sigma}^2\mathbf{I}_q$. In addition, we have

$$\mathbf{S}_t^r = \mathbf{S}_w^r + (1-r)\mathbf{S}_b. \tag{8}$$

Proposition 1. *Let $\mathbf{V}_t$ be the solution of the RLDA problem (5) and $\mathbf{\Lambda}_t = \mathbf{V}_t'\mathbf{S}_b\mathbf{V}_t$. Then the solution $\mathbf{V}_w$ of RLDA problem (7) and $\mathbf{\Lambda}_w = \mathbf{V}_w'\mathbf{S}_b\mathbf{V}_w$ are given by*

$$\mathbf{V}_w = \mathbf{V}_t(\mathbf{I}_q - (1-r)\mathbf{\Lambda}_t)^{-1/2},$$
$$\mathbf{\Lambda}_w = \mathbf{\Lambda}_t(\mathbf{I}_q - (1-r)\mathbf{\Lambda}_t)^{-1}. \tag{9}$$

Proof. Substituting (8) into $\mathbf{V}_t'\mathbf{S}_t^r\mathbf{V}_t$ and noting $\mathbf{V}_t'\mathbf{S}_b\mathbf{V}_t = \mathbf{\Lambda}_t$, we have $\mathbf{V}_t'\mathbf{S}_w^r\mathbf{V}_t = \mathbf{I}_q - (1-r)\mathbf{\Lambda}_t$. Define $\mathbf{V}_w$ as in (9), we obtain $\mathbf{V}_w'\mathbf{S}_w^r\mathbf{V}_w = \mathbf{I}_q$. Furthermore, using (9), we obtain $\mathbf{\Lambda}_w = \mathbf{V}_w'\mathbf{S}_b\mathbf{V}_w = \mathbf{\Lambda}_t(\mathbf{I}_q - (1-r)\mathbf{\Lambda}_t)^{-1}$. This completes the proof. □

Proposition 1 shows the two problems are equivalent, and $\mathbf{V}_w$ and $\mathbf{V}_t$ can be mutually transformed through a diagonal transformation. Due to the diagonal transformation $(\mathbf{I}_q - (1-r)\mathbf{\Lambda}_t)^{-1/2}$ in (9), the solutions $\mathbf{V}_w$ and $\mathbf{V}_t$ are different, and the Euclidean distance between two data points $\mathbf{x}_1$ and $\mathbf{x}_2$ in the low-dimensional discriminant subspace are not equal, namely $||\mathbf{V}_t'(\mathbf{x}_1 - \mathbf{x}_2)|| \neq ||\mathbf{V}_w'(\mathbf{x}_1 - \mathbf{x}_2)||$, which makes the subsequent classifiers such as nearest neighbors using $\mathbf{V}_w$ and $\mathbf{V}_t$ may yield different classification performances.

2.4. Efficient implementation of RLDA with a fixed value of r

Zhang et al. [18] present an efficient implementation of RLDA problem (5) with a fixed value of r, which is briefly reviewed below.

Let $\mathbf{1}_d$ stand for the $d \times 1$ vector of ones, and $\mathbf{E} = (e_{ij})$ be a $n \times c$ indicator matrix with $e_{ij} = 1$ if $\mathbf{x}_i$ belonging to class j and $e_{ij} = 0$ otherwise. The $n \times n$ centering matrix $\mathbf{H} = \mathbf{I}_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n'$.

Moreover, denote $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{R}^{d\times n}$, $\mathbf{M} = [\mathbf{m}_1, \dots, \mathbf{m}_c] \in \mathbb{R}^{d\times c}$, $\mathbf{\Pi} = \mathrm{diag}(n_1, \dots, n_c) \in \mathbb{R}^{c\times c}$, $\mathbf{\Pi}^{1/2} = \mathrm{diag}(\sqrt{n_1}, \dots, \sqrt{n_c})$, $\boldsymbol{\pi} = (n_1, \dots, n_c)' \in \mathbb{R}^{c\times 1}$, $\sqrt{\boldsymbol{\pi}} = (\sqrt{n_1}, \dots, \sqrt{n_c})'$.

Without loss of generality, we assume that the data has been centered, i.e., $\mathbf{X} = \mathbf{XH}$. With the above notations, $\mathbf{S}_b$ in (1) and $\mathbf{S}_t$ in (2) can be rewritten as

$$\mathbf{S}_t = \mathbf{XX}', \tag{10}$$
$$\mathbf{S}_b = \mathbf{F}_b\mathbf{F}_b', \tag{11}$$

where $\mathbf{F}_b = \mathbf{XE\Pi}^{-1/2}$. Substituting (10) and (11) into (6), we obtain

$$\mathbf{GF}_b'\mathbf{V} = \mathbf{V\Lambda},$$

where

$$\mathbf{G} = ((1-r)\mathbf{XX}' + r\hat{\sigma}^2\mathbf{I}_d)^{-1}\mathbf{F}_b. \tag{12}$$

Since $((1-r)\mathbf{XX}' + r\hat{\sigma}^2\mathbf{I}_d)^{-1}\mathbf{X} = \mathbf{X}((1-r)\mathbf{X}'\mathbf{X} + r\hat{\sigma}^2\mathbf{I}_n)^{-1}$, we also have

$$\mathbf{G} = \mathbf{X}((1-r)\mathbf{X}'\mathbf{X} + r\hat{\sigma}^2\mathbf{I}_n)^{-1}\mathbf{E\Pi}^{-1/2}. \tag{13}$$

When $n < d$, it is cheaper to compute $\mathbf{G}$ by (13). Let

$$\mathbf{R} = \mathbf{F}_b'\mathbf{G} = \mathbf{F}_b'((1-r)\mathbf{XX}' + r\hat{\sigma}^2\mathbf{I}_d)^{-1}\mathbf{F}_b. \tag{14}$$

Since the $c \times c$ matrix $\mathbf{R}$ and $d \times d$ matrix $\mathbf{GF}_b'$ have the same nonzero eigenvalues, if $(\mathbf{\Lambda}, \mathbf{V}_R)$ is the eigenpair of $\mathbf{R}$, then we have that $(\mathbf{\Lambda}, \mathbf{GV}_R)$ is the eigenpair of $\mathbf{GF}_b'$. The eigenpair of $\mathbf{R}$ can be obtained by SVD, which is also equivalent to eigenvalue decomposition (EVD) since $\mathbf{R}$ is a nonnegative definite matrix. For clarity, the algorithm for RLDA is summarized in Algorithm 1.

Algorithm 1 Efficient implementation of RLDA for a fixed value of r.

Input: $(\mathbf{X}, \mathbf{E}, \mathbf{\Pi}, r)$.
1: Compute $\mathbf{G}$ by (12) or (13) and $\mathbf{R}$ by (14).
2: Perform the condensed SVD of $\mathbf{R}$ as $\mathbf{R} = \mathbf{V}_R\mathbf{\Lambda}\mathbf{V}_R'$.
Output: $\mathbf{V}_w = \mathbf{GV}_R[\mathbf{\Lambda}(\mathbf{I}_q - (1-r)\mathbf{\Lambda})]^{-1/2}$.

The complexity analysis of Algorithm 1 is given below. Calculating $\mathbf{G}$ costs $O(nda) + O(a^3)$, where $a = \min(n, d)$. Computing $\mathbf{R}$ takes $O(dc^2)$, its eigen-decomposition $O(c^3)$ and $\mathbf{GV}$ $O(dc^2)$. Thus the total cost is $O(nda) + O(dc^2)$. It can be seen that Algorithm 1 is more efficient than the direct implementation in Sec. 2.2, particularly when $d \gg n$. In Sec. 3, we will extend this algorithm to our proposed two-parameter RBLDA.

2.5. Efficient model selection algorithm of RLDA for a candidate set of r

In practice, the regularization parameter r needs to be determined using training data. Usually, to obtain satisfy performance, a large set consisting of m candidates of r is used. The computational cost for the set by the implementation [18] is hence m times higher, namely $O(m(nda + dc^2))$.

Another efficient implementation of RLDA based on SVD is also given in [19]. Its cost for a fixed r is similar to that in [18]. Importantly, based on this implementation, an efficient model selection algorithm is further developed in [19]. The cost of this algorithm for m candidates of r could be comparable with the cost of a fixed r, rather than m times higher. In Sec. 3.3, we will extend the main idea of this algorithm and propose a new model selection algorithm for the two-parameter RBLDA.

2.6. Review of matrix-based discriminant analysis (DA) methods

2.6.1. Bilinear DA (BLDA)

Instead of a linear transformation, bilinear DA (BLDA) seeks for a bilinear transformation $\mathbf{Y} = \mathbf{V}_1'\mathbf{X}\mathbf{V}_2$. The idea originates from bilinear PCA [3]. To find the column and row transformations, a separate solution is presented in [4,21,5]. A brief review of the implementation in [21] is given below.

Given a set of $d_1 \times d_2$-dimensional matrix-valued data $\{\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n\}$ consisting of c classes, let $\mathbf{W} = \frac{1}{n}\sum_i \mathbf{X}_i$ and $\mathbf{W}_k = \frac{1}{n_k}\sum_{i\in\mathcal{L}_k}\mathbf{X}_i$ be the global sample mean and the sample mean of class k, respectively. The column-column and row-row within-class and between-class scatter matrices are defined as

$$\mathbf{S}_{1w} = \frac{1}{nd_2}\sum\nolimits_k \sum\nolimits_{i\in\mathcal{L}_k}(\mathbf{X}_i - \mathbf{W}_k)(\mathbf{X}_i - \mathbf{W}_k)',$$

$$\mathbf{S}_{2w} = \frac{1}{nd_1}\sum\nolimits_k \sum\nolimits_{i\in\mathcal{L}_k}(\mathbf{X}_i - \mathbf{W}_k)'(\mathbf{X}_i - \mathbf{W}_k),$$

$$\mathbf{S}_{1b} = \frac{1}{nd_2}\sum\nolimits_k n_k(\mathbf{W}_k - \mathbf{W})(\mathbf{W}_k - \mathbf{W})', \tag{15}$$

$$\mathbf{S}_{2b} = \frac{1}{nd_1}\sum\nolimits_k n_k(\mathbf{W}_k - \mathbf{W})'(\mathbf{W}_k - \mathbf{W}). \tag{16}$$

It can be easily verified that the ranks of these matrices satisfy $\mathrm{rank}(\mathbf{S}_{1w}) \le \min(d_1, d_2(n-c))$, $\mathrm{rank}(\mathbf{S}_{2w}) \le \min(d_2, d_1(n-c))$, $\mathrm{rank}(\mathbf{S}_{1b}) \le \min(d_1, d_2(c-1))$ and $\mathrm{rank}(\mathbf{S}_{2b}) \le \min(d_1(c-1), d_2)$.

The Fisher criterion for BLDA can be formulated as two separate Fisher sub-criterions [5]

$$\arg\max\nolimits_{\mathbf{V}_{lw}} \mathrm{tr}\left\{(\mathbf{V}_{lw}'\mathbf{S}_{lw}\mathbf{V}_{lw})^{-1}(\mathbf{V}_{lw}'\mathbf{S}_{lb}\mathbf{V}_{lw})\right\},$$

$l = 1, 2$, each of which is a similar optimization problem to that in LDA, with $l = 1$ devoted to column direction and $l = 2$ to row direction. The solution is obtained by solving the two problems separately

$$\max\nolimits_{\mathbf{V}_{lw}} \mathrm{tr}\left\{\mathbf{V}_{lw}'\mathbf{S}_{lb}\mathbf{V}_{lw}\right\},\ s.t.\ \mathbf{V}_{lw}'\mathbf{S}_{lw}\mathbf{V}_{lw} = \mathbf{I}, \tag{17}$$

where $l = 1, 2$. Assume that $\mathbf{S}_{lw}$ is invertible and let the EVD of $\mathbf{S}_{lw}$ be

$$\mathbf{S}_{lw} = \mathbf{U}_{lw}\mathbf{\Gamma}_{lw}\mathbf{U}_{lw}'.$$

Denote $\widetilde{\mathbf{V}}_{lw} = \mathbf{\Gamma}_{lw}^{1/2}\mathbf{U}_{lw}'\mathbf{V}_{lw}$. Substitute $\mathbf{V}_{lw} = \mathbf{U}_{lw}\mathbf{\Gamma}_{lw}^{-1/2}\widetilde{\mathbf{V}}_{lw}$ into (17), yielding

$$\max\nolimits_{\widetilde{\mathbf{V}}_{lw}} \mathrm{tr}\left\{\widetilde{\mathbf{V}}_{lw}'\mathbf{R}_l\widetilde{\mathbf{V}}_{lw}\right\},\ s.t.\ \widetilde{\mathbf{V}}_{lw}'\widetilde{\mathbf{V}}_{lw} = \mathbf{I}, \tag{18}$$

where $\mathbf{R}_l = \mathbf{\Gamma}_{lw}^{-1/2}\mathbf{U}_{lw}'\mathbf{S}_{lb}\mathbf{U}_{lw}\mathbf{\Gamma}_{lw}^{-1/2}$. Denote the solution to problem (18) be $\mathbf{V}_{lR}$, the solution to problem (17) is then given by $\mathbf{V}_{lw} = \mathbf{U}_{lw}\mathbf{\Gamma}_{lw}^{-1/2}\mathbf{V}_{lR}$.

Next we analyze the main time complexity of this procedure in column direction since the analysis in row direction is similar. The formation of the $d_1 \times d_1$ matrix $\mathbf{S}_{1w}$ costs $O(d_1^2 d_2 n)$ and its eigen-decomposition takes $O(d_1^3)$. Computing $\mathbf{R}_1$ and its eigen-decomposition cost $O(d_1^3)$. Hence the total cost for $\mathbf{V}_{1w}$ is $O(d_1^2 d_2 n) + O(d_1^3)$. Thus this cost could be very heavy when d_1 is much greater than d_2. This is one shortcoming of this procedure. Another shortcoming is that it requires both of $\mathbf{S}_{1w}$ and $\mathbf{S}_{2w}$ to be invertible. Since $\mathrm{rank}(\mathbf{S}_{1w}) \le \min(d_1, d_2(n-c))$ and $\mathrm{rank}(\mathbf{S}_{2w}) \le \min(d_2, d_1(n-c))$, for data $d_1 > d_2(n-c)$ or $d_2 > d_1(n-c)$, $\mathbf{S}_{1w}$ or $\mathbf{S}_{2w}$ is singular and thus this procedure cannot be performed. Motivated by the success of RLDA, in Sec. 3 we extend the idea of RLDA to BLDA for overcoming the two shortcomings.

2.6.2. Related matrix-based DA methods on MTS data

The assumption of separable transformation has been successfully explored and demonstrated its effectiveness in the discriminant analysis of MTS data. Various approaches have been proposed to address this issue.

For example, the bilinear support vector machine (SVM) proposed in [22] specifically designed for spatio-temporal visual data. In the context of EEG data analysis, two different bilinear discriminant analysis (DA) methods have been proposed: one in [23] and another in [24], both of which are based on logistic regression. One significant advantage of these bilinear DA methods compared to their linear counterparts is their ability to simultaneously provide the interesting spatial and temporal profiles. This advantage stems from the separable transformation employed in these methods. Although matrix-based DA methods, such as the ones mentioned above, have achieved success in MTS data analysis, there has been limited research comparing them with their linear counterparts in terms of classification performance.

3. Two-parameter regularized BLDA

In this section, we propose a new two-parameter regularized BLDA (RBLDA) for MTS data classification. In Sec. 3.1, we formulate the optimization problem for our RBLDA. In Sec. 3.2, we present an efficient implementation. In Sec. 3.3 we propose an efficient model selection algorithm for our two-parameter RBLDA.

3.1. The proposed two-parameter RBLDA

Similar to those in RLDA, we define column-column and row-row total scatter matrices as

$$\mathbf{S}_{1t} = \frac{1}{nd_2}\sum\nolimits_i (\mathbf{X}_i - \mathbf{W})(\mathbf{X}_i - \mathbf{W})', \tag{19}$$

$$\mathbf{S}_{2t} = \frac{1}{nd_1}\sum\nolimits_i (\mathbf{X}_i - \mathbf{W})'(\mathbf{X}_i - \mathbf{W}). \tag{20}$$

We solve the following two eigenproblems,

$$\mathbf{S}_{lt}^{r_l\,-1}\mathbf{S}_{lb}\mathbf{V}_l = \mathbf{V}_l\boldsymbol{\Lambda}_l, \quad l = 1, 2, \tag{21}$$

under the constraints that $\mathbf{V}_l'\mathbf{S}_{lt}\mathbf{V}_l = \mathbf{I}_{q_l}$, where

$$\mathbf{S}_{lt}^{r_l} = (1 - r_l)\mathbf{S}_{lt} + r_l\hat{\sigma}_l^2\mathbf{I}_{d_l}, \tag{22}$$

$\hat{\sigma}_l^2 = \mathrm{tr}(\mathbf{S}_{lt})/d_l$, and the regularization parameters $r_1, r_2 \in (0, 1]$. Note that the solutions $\mathbf{V}_{lw}$ to the optimization problems (21) under the constraints that $\mathbf{V}_{lw}'\mathbf{S}_{lw}\mathbf{V}_{lw} = \mathbf{I}$ can be easily obtained by (9).

3.2. Implementation 1: an efficient implementation of RBLDA with a fixed value of (r_1, r_2)

In this subsection, we extend the efficient implementation of RLDA in Sec. 2.4 to RBLDA and develop an efficient implementation for RBLDA with a fixed value of the regularization parameter (r_1, r_2), which is denoted as Implementation 1 for clarity. We first consider the implementation in column direction with r_1 fixed. Let

$$\begin{aligned}
\mathbf{X}_{(1)} &= [\mathbf{X}_1, \dots, \mathbf{X}_n]\ (d_1 \times d_2 n),\\
\mathbf{W}_{(1)} &= [\mathbf{W}_1, \dots, \mathbf{W}_c]\ (d_1 \times d_2 c),\\
\boldsymbol{\Pi}_1 &= \mathrm{blkdiag}(n_1\mathbf{I}_{d_2}, \dots, n_c\mathbf{I}_{d_2})\ (d_2 c \times d_2 c),\\
\boldsymbol{\Pi}_1^{1/2} &= \mathrm{blkdiag}(\sqrt{n_1}\mathbf{I}_{d_2}, \dots, \sqrt{n_c}\mathbf{I}_{d_2})\ (d_2 c \times d_2 c),\\
\boldsymbol{\pi}_1 &= (n_1\mathbf{I}_{d_2}, \dots, n_c\mathbf{I}_{d_2})'\ (d_2 c \times d_2),\\
\sqrt{\boldsymbol{\pi}_1} &= (\sqrt{n_1}\mathbf{I}_{d_2}, \dots, \sqrt{n_c}\mathbf{I}_{d_2})'\ (d_2 c \times d_2),\\
\mathbb{I}_{1m} &= (\mathbf{I}_{d_2}, \dots, \mathbf{I}_{d_2})'\ (d_2 m \times d_2),\\
\mathbf{H}_1 &= \mathbf{I}_{d_2 n} - \frac{1}{n}\mathbb{I}_{1n}\mathbb{I}_{1n}'\ (d_2 n \times d_2 n),\\
\mathbf{H}_{1\pi} &= \mathbf{I}_{d_2 c} - \frac{1}{n}\sqrt{\boldsymbol{\pi}_1}\sqrt{\boldsymbol{\pi}_1}'\ (d_2 c \times d_2 c),
\end{aligned}$$

$\mathbb{E}_1 = (\mathbf{E}_{ij})\ (d_2 n \times d_2 c)$ be a $n \times c$ partitioned matrix with the $d_2 \times d_2$ ij-th submatrices $\mathbf{E}_{ij} = \mathbf{I}_{d_2}$ if $\mathbf{X}_i$ belonging to class j and $\mathbf{E}_{ij} = \mathbf{0}_{d_2}$ otherwise.

Without loss of generality, we assume that the data has been centered, i.e., $\mathbf{X}_{(1)} = \mathbf{X}_{(1)}\mathbf{H}_1$. With these representations, we have $\mathbb{I}_{1n}'\mathbb{E}_1 = \mathbb{I}_{1c}'\boldsymbol{\Pi}_1 = \boldsymbol{\pi}_1'$, $\mathbb{E}_1\mathbb{I}_{1c} = \mathbb{I}_{1n}$, $\mathbb{I}_{1c}'\boldsymbol{\pi}_1 = n\mathbf{I}_{d_2}$, $\mathbb{E}_1'\mathbb{E}_1 = \boldsymbol{\Pi}_1$, $\boldsymbol{\Pi}_1^{-1}\boldsymbol{\pi}_1 = \mathbb{I}_{1c}$, and

$$\begin{aligned}
\mathbf{W}_{(1)} &= \mathbf{X}_{(1)}\mathbb{E}_1\boldsymbol{\Pi}_1^{-1},\\
\mathbb{E}_1\boldsymbol{\Pi}_1^{-1/2}\mathbf{H}_{1\pi} &= \mathbb{E}_1\boldsymbol{\Pi}_1^{-1/2} - \frac{1}{n}\mathbb{I}_{1n}\sqrt{\boldsymbol{\pi}_1}' = \mathbf{H}_1\mathbb{E}_1\boldsymbol{\Pi}_1^{-1/2},
\end{aligned}$$

Noting that $\mathbf{H}_1 = \mathbf{H}_1^2$, $\mathbf{S}_{1t}$ in (19) can be rewritten as

$$\mathbf{S}_{1t} = \mathbf{X}_{(1)}\mathbf{X}_{(1)}', \tag{23}$$

and $\mathbf{S}_{1b}$ in (15) as

$$\begin{aligned}
\mathbf{S}_{1b} &= \mathbf{W}_{(1)}\left[\boldsymbol{\Pi}_1 - \frac{1}{n}\boldsymbol{\pi}_1\boldsymbol{\pi}_1'\right]\mathbf{W}_{(1)}'\\
&= \mathbf{W}_{(1)}\left[\boldsymbol{\Pi}_1^{1/2} - \frac{1}{n}\boldsymbol{\pi}_1\sqrt{\boldsymbol{\pi}_1}'\right]\left[\boldsymbol{\Pi}_1^{1/2} - \frac{1}{n}\sqrt{\boldsymbol{\pi}_1}\boldsymbol{\pi}_1'\right]\mathbf{W}_{(1)}'
\end{aligned}$$

$$
\begin{aligned}
&= \mathbf{X}_{(1)}\mathbb{E}_1\mathbf{\Pi}_1^{-1}\mathbf{\Pi}_1^{1/2}\mathbf{H}_{1\pi}\mathbf{H}_{1\pi}\mathbf{\Pi}_1^{1/2}\mathbf{\Pi}_1^{-1}\mathbb{E}_1\mathbf{X}'_{(1)}\\
&= \mathbf{X}_{(1)}\mathbb{E}_1\mathbf{\Pi}_1^{-1}\mathbb{E}'_1\mathbf{X}'_{(1)}\\
&= \mathbf{F}_{1b}\mathbf{F}'_{1b},
\end{aligned}
\tag{24}
$$

where $\mathbf{F}_{1b} = \mathbf{X}_{(1)}\mathbb{E}_1\mathbf{\Pi}_1^{-1/2}$. In addition, we define

$$
\begin{aligned}
\mathbf{T}_1 &= \mathbf{X}'_{(1)}\mathbf{X}_{(1)},\\
\mathbf{T}_1^{r_1} &= (1-r_1)\mathbf{X}'_{(1)}\mathbf{X}_{(1)} + r_1\hat{\sigma}_1^2\mathbf{I}_{d_2 n}
\end{aligned}
\tag{25}
$$

Let the condensed SVD of $\mathbf{S}_{1t}(d_1 \times d_1)$ in (23) and $\mathbf{T}_1(d_2 n \times d_2 n)$ be

$$
\mathbf{S}_{1t} = \mathbf{X}_{(1)}\mathbf{X}'_{(1)} = \mathbf{U}_{1t}\mathbf{\Gamma}_{1t}\mathbf{U}'_{1t},
\tag{26}
$$

$$
\mathbf{T}_1 = \mathbf{X}'_{(1)}\mathbf{X}_{(1)} = \mathbf{U}_{1T}\mathbf{\Gamma}_{1t}\mathbf{U}'_{1T},
\tag{27}
$$

where $t_1 = \mathrm{rank}(\mathbf{\Gamma}_{1t})$. When $d_1 > d_2 n$, the condensed SVD in (26) can be obtained via (27). Thus in this case the complexity of the condensed SVD of $\mathbf{S}_{1t}$ is $O(d_1(d_2 n)^2)$.

Accordingly, the EVD of $\mathbf{S}_{1t}^{r_1}$ in (22) and $\mathbf{T}_1^{r_1}$ (25) are given by

$$
\begin{aligned}
\mathbf{S}_{1t}^{r_1} &= \mathbf{U}_{1t}\mathbf{\Gamma}_{1t}^{r_1}\mathbf{U}'_{1t} + r_1\hat{\sigma}_1^2\mathbf{U}_{1t}^{\perp}\mathbf{U}_{1t}^{\perp\prime},\\
\mathbf{T}_1^{r_1} &= \mathbf{U}_{1T}\mathbf{\Gamma}_{1T}^{r_1}\mathbf{U}'_{1T} + r_1\hat{\sigma}_1^2\mathbf{U}_{1T}^{\perp}\mathbf{U}_{1T}^{\perp\prime},
\end{aligned}
\tag{28}
$$

where $\mathbf{\Gamma}_{1t}^{r_1} = (1-r_1)\mathbf{\Gamma}_{1t} + r_1\hat{\sigma}_1^2\mathbf{I}_{d_1}$ $\mathbf{\Gamma}_{1T}^{r_1} = (1-r_1)\mathbf{\Gamma}_{1T} + r_1\hat{\sigma}_1^2\mathbf{I}_{d_2 n}$, and $\mathbf{U}_{1t}^{\perp}$ is the orthogonal complement of $\mathbf{U}_{1t}$.

Next, we solve the eigenproblem in (21) in two cases: $d_2 c < d_1$ and $d_2 c \geq d_1$, under which we will see that the problem can be implemented with different computational costs.

(i) When $d_2 c \geq d_1$, with (28), the eigenproblem in (21) is

$$
\mathbf{U}_{1t}\mathbf{\Gamma}_{1t}^{r_1\,-1/2}\mathbf{G}_{11}\mathbf{V}_1 = \mathbf{V}_1\mathbf{\Lambda}_1,
$$

where

$$
\mathbf{G}_{11} = \mathbf{\Gamma}_{1t}^{r_1\,-1/2}\mathbf{U}'_{1t}\mathbf{S}_{1b}.
$$

Let

$$
\begin{aligned}
\mathbf{R}_{11} &= \mathbf{G}_{11}\mathbf{U}_{1t}\mathbf{\Gamma}_{1t}^{r_1\,-1/2}\\
&= \mathbf{\Gamma}_{1t}^{r_1\,-1/2}\mathbf{U}'_{1t}\mathbf{S}_{1b}\mathbf{U}_{1t}\mathbf{\Gamma}_{1t}^{r_1\,-1/2}.
\end{aligned}
\tag{29}
$$

Since the $d_1 \times d_1$ matrices $\mathbf{R}_{11}$ and $\mathbf{U}_{1t}\mathbf{\Gamma}_{1t}^{r_1\,-1/2}\mathbf{G}_{11}$ have the same nonzero eigenvalues, if $(\mathbf{\Lambda}_1, \mathbf{V}_{1R})$ is the eigenpair of $\mathbf{R}_{11}$ and $\mathbf{V}_1 = \mathbf{U}_{1t}\mathbf{\Gamma}_{1t}^{r_1\,-1/2}\mathbf{V}_{1R}$, we have $(\mathbf{\Lambda}_1, \mathbf{V}_1)$ is the eigenpair of $\mathbf{U}_{1t}\mathbf{\Gamma}_{1t}^{r_1\,-1/2}\mathbf{G}_{11}$.

(ii) When $d_2 c < d_1$, using (24), the eigenproblem in (21) can be represented by

$$
\mathbf{G}_{12}\mathbf{F}'_{1b}\mathbf{V}_1 = \mathbf{V}_1\mathbf{\Lambda}_1,
$$

where

$$
\mathbf{G}_{12} = \mathbf{S}_{1t}^{r_1\,-1}\mathbf{F}_{1b}\ (d_1 \times d_2 c).
\tag{30}
$$

Since $\mathbf{S}_{1t}^{r_1\,-1}\mathbf{X}_{(1)} = \mathbf{X}_{(1)}\mathbf{T}_1^{r_1\,-1}$, we also have

$$
\mathbf{G}_{12} = \mathbf{X}_{(1)}\mathbf{T}_1^{r_1\,-1}\mathbb{E}_1\mathbf{\Pi}_1^{-1/2}.
\tag{31}
$$

When $d_2 n < d_1$, it is cheaper to compute $\mathbf{G}_{12}$ by (31). Let

$$
\begin{aligned}
\mathbf{R}_{12} &= \mathbf{\Pi}_1^{-1/2}\mathbb{E}'_1\mathbf{X}'_{(1)}\mathbf{G}_{12}\ (d_2 c \times d_2 c)\\
&= \mathbf{F}'_{1b}\mathbf{S}_{1t}^{r_1\,-1}\mathbf{F}_{1b}.
\end{aligned}
\tag{32}
$$

The $d_2 c \times d_2 c$ matrix $\mathbf{R}_{12}$ and $d_1 \times d_1$ matrix $\mathbf{G}_{12}\mathbf{F}'_{1b}$ have the same nonzero eigenvalues, if $(\mathbf{\Lambda}_1, \mathbf{V}_{1R})$ is the eigenpair of $\mathbf{R}_{12}$ and $\mathbf{V}_1 = \mathbf{G}_{12}\mathbf{V}_{1R}$, $(\mathbf{\Lambda}_1, \mathbf{V}_1)$ is the eigenpair of $\mathbf{G}_{12}\mathbf{F}'_{1b}$.

When $d_2 c \geq d_1$, the cost of Implementation 1 is the same as that of BLDA in Sec. 2.6.1, i.e., $O(d_1^2 d_2 n) + O(d_1^3)$. When $d_2 c < d_1$, calculating $\mathbf{G}_{12}$ via (30) or (31) costs $O(d_1 d_2 n a_1) + O(d_1 d_2 c a_1)$, where $a_1 = \min(d_1, d_2 n)$. Computing $\mathbf{R}_{12}$ and $\mathbf{G}_{12}\mathbf{V}_1$ takes $O(d_1(d_2 c)^2)$. Feature extraction for $\mathbf{X}$ (i.e., computation of $\mathbf{V}'_1\mathbf{X}_{(1)}$) takes $O(d_1 d_2 n d_2 c)$. Thus the total cost for a single value of r_1 is $O(d_1 d_2 n a_1 + d_1 d_2 n d_2 c)$. Compared with that of BLDA, the cost with this implementation is substantially reduced. However, if a total of m_1

candidates of r_1 are considered, then the total cost of running Implementation 1 would be m_1 times higher, i.e., $O(m_1(d_1d_2na_1 + d_1d_2nd_2c))$.

The computation in row direction with r_2 fixed is similar to that in column direction. The details are omitted here as we will develop a variant of Implementation 1 in Sec. 3.3.1.

3.3. Efficient model selection

To choose regularization parameter (r_1, r_2), in this subsection we propose an efficient model selection algorithm of RBLDA. In Sec. 3.3.1, we propose a variant of Implementation 1 in Sec. 3.2, which is more efficient for multiple candidates. Based on Implementation 2, we give an efficient model selection algorithm in Sec. 3.3.2.

3.3.1. Implementation 2: a variant of Implementation 1 for multiple candidates of (r_1, r_2)

To further reduce costs employing the implementation in Sec. 3.2 for multiple candidates of (r_1, r_2), we propose a variant in this subsection. For clarity, this variant is denoted by Implementation 2. The main advantage of Implementation 2 is that it enables the implementations for multiple candidates to share certain common computations. As a result, the cost of running Implementation 2 for multiple candidates can be significantly reduced compared to Implementation 1 in Sec. 3.2.

For the implementation in column direction, let

$$\mathbf{X}_{(1u)} = \mathbf{U}'_{1t}\mathbf{X}_{(1)}\ (t_1 \times d_2 n), \tag{33}$$

where $t_1 = \text{rank}(\mathbf{X}_{(1)})$. By (28), the regularized total scatter matrix in $\mathbf{X}_{(1u)}$-space is

$$\mathbf{S}^{r_1}_{1tu} = \mathbf{\Gamma}^{r_1}_{1t}. \tag{34}$$

Define

$$\mathbf{F}_{1bu} = \mathbf{X}_{(1u)}\mathbb{E}_1\mathbf{\Pi}_1^{-1/2}\ (t_1 \times d_2 c). \tag{35}$$

(i) When $d_2c \geq d_1$, with (35), $\mathbf{R}_{11}$ (29) in $\mathbf{X}_{(1u)}$-space remains unchanged, i.e.,

$$\mathbf{R}_{11} = \mathbf{\Gamma}^{r_1-1/2}_{1t}\mathbf{F}_{1bu}\mathbf{F}'_{1bu}\mathbf{\Gamma}^{r_1-1/2}_{1t}, \tag{36}$$

and the $\mathbf{V}$-solution in $\mathbf{X}_{(1u)}$-space is $\mathbf{V}_{1u} = \mathbf{\Gamma}^{r_1-1/2}_{1t}\mathbf{V}_{1R}(t_1 \times d_2c)$.
(ii) When $d_2c < d_1$, by (28), $\mathbf{G}_{12}$ and $\mathbf{R}_{12}$ in $\mathbf{X}_{(1u)}$-space ((30) and (32)) are

$$\mathbf{G}_{12u} = \mathbf{\Gamma}^{r_1-1}_{1t}\mathbf{F}_{1bu}\ (t_1 \times d_2c), \tag{37}$$

$$\mathbf{R}_{12} = \mathbf{F}'_{1bu}\mathbf{G}_{12u} = \mathbf{F}'_{1bu}\mathbf{\Gamma}^{r_1-1}_{1t}\mathbf{F}_{1bu}, \tag{38}$$

and the $\mathbf{V}$-solution in $\mathbf{X}_{(1u)}$-space is $\mathbf{V}_{1u} = \mathbf{G}_{12u}\mathbf{V}_{1R}\ (t_1 \times d_2c)$. It will be seen from Sec. 3.3.2 that when d_1 is much greater than d_2, the cost of Implementation 2 in obtaining $\mathbf{V}_{1u}$ is $O(d_1d_2na_1 + d_2nd_2ca_1)$. Since $a_1 \leq d_1$, the cost could be slightly reduced compared with Implementation 1 in Sec. 3.2.

The $\mathbf{V}$-solution in the original $\mathbf{X}_{(1)}$-space can be obtained by $\mathbf{V}_1 = \mathbf{U}_{1t}\mathbf{V}_{1u}$ under both cases. If obtaining $\mathbf{V}_1$ is necessary, then the resulting cost would remain the same as Implementation 1 in Sec. 3.2. However, when using the nearest neighbor classifier, it is not necessary to obtain $\mathbf{V}_1$ since the distance computations based on $\mathbf{V}_{1u}$ and $\mathbf{V}_1$ are equivalent. Since the computation in row direction with r_2 fixed is similar to that in column direction, the details are provided in Appendix A.

Thanks to (33), which is the same for m_1 candidates of r_1, and hence the cost of Implementation 2 in Sec. 3.3.1 for m_1 candidates of r_1 can be reduced to $O(d_1d_2na_1 + m_1(d_2nd_2ca_1))$ only.

3.3.2. Efficient model selection algorithm of RBLDA for a set of (r_1, r_2)

With Implementation 1, cross validation (CV) can be employed to choose the regularization parameter from all m_1m_2 candidates of (r_1, r_2). In the traditional v-fold CV procedure, Implementation 1 needs to be applied separately to each of the v folds. On each fold, Implementation 1 has to be run a total of m_1m_2 times for all the candidate values of (r_1, r_2). Thus the traditional v-fold CV is expensive.

Building upon the Implementation 2 in Sec. 3.3.1, the cross validation for RBLDA can now be performed efficiently. For clarity, the complete model selection algorithm of RBLDA for a set of (r_1, r_2) is summarized in Algorithm 2.

Below we analyze the complexity of the feature extraction stage of Algorithm 2 (i.e., not involving the classification stage (Line 23)). Line 3 takes $O(d_1d_2na_1)$ and line 4 costs $O(t_1d_1d_2n)$. When $d_2c \geq d_1$, line 7 takes $O(t_1^2d_2c)$ and line 9 $O(t_1^3)$, and line 8 and 9 take $O(t_1(d_2c)^2)$ otherwise. Line 20 and Line 22 take $O(t_1d_2n \cdot \min(t_1, d_2c))$. The total cost for m_1 and m_2 candidates in column and row directions is

$$\begin{aligned} T(m_1, m_2) = O\big(& d_1d_2na_1 + m_1m_2(t_2d_2\min(t_1, d_2c)n) \\ & + m_1(t_1d_2c + t_1d_2n)\min(t_1, d_2c)\big). \end{aligned}$$

Algorithm 2 Efficient RBLDA model selection algorithm.

Input: $(\mathbf{X}, \mathbb{E}_l, \mathbf{\Pi}_l, \{r_{li}\}_{i=1}^{m_l}, l = 1, 2)$.
for $v = 1 : V$ **do**
Split $\mathbf{X}$ as training $\mathbf{X}_v$ and validation $\mathbf{X}_v^{\perp}$;
Perform condensed SVD of $\mathbf{S}_{1t}$ by (26) on $\mathbf{X}_v$.
Compute $\mathbf{X}_{(1u)}$ by (33), $\mathbf{F}_{1bu}$ by (35) on $\mathbf{X}_v$.
for $i = 1 : m_1$ **do**
Given r_{1i}. // i-th candidate from m_1 choices of r_1.
When $d_2 c \geq d_1$, compute $\mathbf{R}_{11}$ by (36), and
compute $\mathbf{G}_{12u}$ by (37) and $\mathbf{R}_{12}$ by (38) otherwise.
Perform the condensed SVD of $\mathbf{R}_{11}$ or $\mathbf{R}_{12}$ as $\mathbf{V}_{1R}\mathbf{\Lambda}_1\mathbf{V}'_{1R}$ and set $\mathbf{V}_{1ui} = \mathbf{\Gamma}_{1t}^{r_1 - 1/2}\mathbf{V}_{1R}$ or $\mathbf{V}_{1ui} = \mathbf{G}_{12u}\mathbf{V}_{1R}[\mathbf{\Lambda}_1(\mathbf{I} - (1 - r_1)\mathbf{\Lambda}_1)]^{-\frac{1}{2}}$.
end for
Perform condensed SVD of $\mathbf{S}_{2t}$ by (A.2) on $\mathbf{X}_v$.
Compute $\mathbf{X}_{(2u)}$ by (A.4), $\mathbf{F}_{2bu}$ by (A.5) on $\mathbf{X}_v$.
for $j = 1 : m_2$ **do**
Given r_{2j}. // j-th candidate from m_2 choices of r_2
When $d_1 c \geq d_2$, compute $\mathbf{R}_{21}$ by (A.6), and
compute $\mathbf{G}_{22u}$ by (A.7) and $\mathbf{R}_{22}$ by (A.8) otherwise.
Perform the condensed SVD of $\mathbf{R}_{21}$ or $\mathbf{R}_{22}$ as $\mathbf{V}_{2R}\mathbf{\Lambda}_2\mathbf{V}'_{2R}$ and set $\mathbf{V}_{2uj} = \mathbf{\Gamma}_{2t}^{r_2 - 1/2}\mathbf{V}_{2R}$ or $\mathbf{V}_{2uj} = \mathbf{G}_{22u}\mathbf{V}_{2R}[\mathbf{\Lambda}_2(\mathbf{I} - (1 - r_2)\mathbf{\Lambda}_2)]^{-\frac{1}{2}}$.
end for
for $i = 1 : m_1$ **do**
$\mathbf{X}_v^i \leftarrow \mathbf{V}'_{1ui}\mathbf{X}_v$,
$\mathbf{X}_v^{i\perp} \leftarrow \mathbf{V}'_{1ui}\mathbf{X}_v^{\perp}$.
for $j = 1 : m_2$ **do**
$\mathbf{X}_v^{ij} \leftarrow \mathbf{X}_v^i\mathbf{V}_{2uj}$,
$\mathbf{X}_v^{ij\perp} \leftarrow \mathbf{X}_v^{i\perp}\mathbf{V}_{2uj}$.
Run 1NN on $(\mathbf{X}_v^{ij}, \mathbf{X}_v^{ij\perp})$ and compute the error rate $\mathrm{Err}(v, i, j)$.
end for
end for
end for
$\mathrm{Err}(i, j) = 1/v\sum_{v=1}^{V}\mathrm{Err}(v, i, j)$.
$(i^*, j^*) \leftarrow \arg\min_{(i,j)}\mathrm{Err}(i, j)$.
Output: (r_{1i^*}, r_{2j^*}) as the best regularization parameter.

(i) When $d_2 c > d_1$ and $d_1 c > d_2$, i.e., $d_2 \approx d_1$, the costs in column and row directions are about the same and thus we only analyze the cost in column direction. In this case, $d_2 c > t_1$, $t_1 \leq a_1$ and hence

$$T(m_1, m_2) = O(d_1^2 d_2 n + m_1(d_1^2 d_2 n) + m_1 m_2(d_2^2 d_1 n)).$$

The ratio of $T(m_1, m_2)$ to $T(1, 1)$ can be approximately expressed as

$$\frac{T(m_1, m_2)}{T(1, 1)} \approx \frac{m_1 m_2}{3}.$$

(ii) When $d_1 > d_2 c$, in particular, d_1 is much higher than d_2, the cost mainly lies in column direction. For simplicity, we assume that $t_1 > d_2 c$ and obtain

$$T(m_1, m_2) = O(d_1 d_2 n a_1 + m_1(d_2 n d_2 c a_1) + m_1 m_2(d_2^2 d_2 c n)).$$

Note that for single (r_1, r_2), the cost of $T(1, 1)$ in this case can be expressed as $O(d_1 d_2 n a_1 + d_2 n d_2 c a_1)$, which is mentioned in Sec. 3.3.1. The ratio of $T(m_1, m_2)$ to $T(1, 1)$ can be approximately expressed as

$$\frac{T(m_1, m_2)}{T(1, 1)} \approx 1 + m_1\frac{d_2 c}{d_1} + m_1 m_2\frac{d_2^2 c}{d_1 a_1},$$

which indicates that the proposed Algorithm 2 is efficient when d_1 is much higher than d_2.

A similar conclusion can be drawn from the complexity analysis when d_2 is much higher than d_1. In a word, the proposed Algorithm 2 is efficient when one data dimensionality is much higher than the other, i.e., $d_1 \gg d_2 n$ or $d_2 \gg d_1 n$.

3.3.3. A brief summary of related methods

For clarity, Table 1 provides a comprehensive summary of the similarities and differences among the related methods discussed in the paper. Table 1 makes clearer the main contributions of the proposed two-parameter RBLDA and Algorithm 2.

4. Experiments

In this section, we perform experiments on the following five publicly available real-world MTS data sets.[2] For comparison, we also provide some results on image data in Appendix B.

- AUSLAN dataset (AUS). AUS contains 2565 MTS observations of 95 signs (i.e. 95 classes). Each sign has 27 observations, each captured from a native AUSLAN speaker using 22 sensors (i.e. 22 variables) on the CyberGlove. In our experiments, we use a subset consisting of 675 observations of 25 signs, each 27 observations. The 25 signs are alive, all, boy, building, buy, cold, come, computer, cost, crazy, danger, deaf, different, girl, glove, go, God, joke, juice, man, where, which, yes, you and zero. The time length of MTS observation we use is 47 and hence the data dimension is 47×22.
- ECG dataset. ECG comprises 200 MTS observations of two classes. Each observation is collected by two electrodes (i.e. 2 variables) during one heartbeat and labeled as normal or abnormal. Abnormal heartbeats signal a cardiac pathology known as

[2] AUS and JAP: available from http://archive.ics.uci.edu/ml/datasets, ECG and WAF: http://www.mustafabaydogan.com./files/viewcategory/20-data-sets.html, BCI: http://www.bbci.de/competition/ii/.

Table 1
A brief summary of related methods.

Type	Method	Impl./Algo.	Optimization problem		When one within-class matrix is singular	
			Objective	Constraint	Applicable or not	Computational cost
Vector-based LDA (Linear: $\mathbf{y} = \mathbf{V}'_w \mathbf{x}$)	LDA	direct impl. by (6)	max : $\mathrm{tr}\{\mathbf{V}'_w \mathbf{S}_b \mathbf{V}_w\}$	$\mathbf{V}'_w \mathbf{S}_w \mathbf{V}_w = \mathbf{I}_q$	not applicable	
	RLDA	direct impl. by (6)		single r: $\mathbf{V}'_w \mathbf{S}^r_w \mathbf{V}_w = \mathbf{I}_q$	applicable	cost1: $O(d^2 n) + O(d^3)$.
		impl. in [20]			applicable	cost2: $O(nda) + O(dc^2)$, cheaper than cost1 for a fixed r. cost3 (cost for m candidates of r): m times cost2.
		model selection algo. in [19]		m candidate set $\{r_i\}_{i=1}^m$: $\mathbf{V}'_w \mathbf{S}^{r_i}_w \mathbf{V}_w = \mathbf{I}_q$	applicable	cost4 (cost for m candidates of r): could be close cost2; much cheaper than cost3 for a set of r.
Matrix-based LDA (Bilinear: $\mathbf{Y} = \mathbf{V}'_{1w} \mathbf{X} \mathbf{V}_{2w}$)	BLDA [4]	direct impl. by (18)	max: $\mathrm{tr}\{\mathbf{V}'_{lw} \mathbf{S}_{lb} \mathbf{V}_{lw}\}$, $l = 1, 2$	$\mathbf{V}'_{lw} \mathbf{S}_{lw} \mathbf{V}_{lw} = \mathbf{I}_{q_l}$	not applicable	
	PBLDA [16]	impl. in [16]			applicable	cost5: cheaper than cost6.
	one-param. RBLDA [7]	direct impl. by (18)		fixed r ($r_1 = r_2$): $\mathbf{V}'_{lw} \mathbf{S}^r_{lw} \mathbf{V}_{lw} = \mathbf{I}_{q_l}$	applicable	cost6 (cost for a fixed (r_1, r_2)): $O(d_1^2 d_2 n) + O(d_2^2 d_1 n) + O(d_1^3) + O(d_2^3)$. cost7 (cost for m candidates of r): m times cost6.
	New: two-param. RBLDA	direct impl. by (18)		fixed (r_1, r_2): $\mathbf{V}'_{lw} \mathbf{S}^{r_l}_{lw} \mathbf{V}_{lw} = \mathbf{I}_{q_l}$	applicable	cost8 (cost for a fixed (r_1, r_2)): cost6. cost9 (cost for $m_1 m_2$ candidates of (r_1, r_2)): $m_1 m_2$ times cost6.
	New: two-param. RBLDA	Algo. 2 in this paper		$m_1 m_2$ candidate set $\{(r_{1i}, r_{2j})\}_{i=1,j=1}^{m_1,m_2}$: $\mathbf{V}'_{lw} \mathbf{S}^{r_l}_{lw} \mathbf{V}_{lw} = \mathbf{I}_{q_l}$	applicable	cost10 (cost for $m_1 m_2$ candidates of (r_1, r_2)): could be close cost5; much cheaper than cost9 for a set of (r_1, r_2).

Table 2
Several statistics for the five MTS data sets. n: Total number of observations; $d_1 \times d_2$: matrix observation size; c: number of classes; p_1, and p_2: small and large training proportions; $B_{tr} = 316/416$.

Dataset	$d_1 \times d_2 \times n$	c	p_1	p_2
AUSLAN	$47 \times 22 \times 675$	25	1/9	1/4
ECG	$39 \times 2 \times 200$	2	1/10	4/5
JAPAN	$7 \times 12 \times 640$	9	1/20	4/5
WAFER	$104 \times 6 \times 327$	2	1/4	4/5
BCI	$500 \times 28 \times 316$	2	$1/16 \cdot B_{tr}$	B_{tr}

supraventricular premature beat. The normal and abnormal classes have 133 and 67 observations, respectively. The time length used is 39 and thus the observation size is 39×2.

- Japanese vowels dataset (JAP). JAP contains 640 MTS observations of nine male speakers. Each observation is the utterance of two Japanese Vowels /ae/ from a speaker recorded by 12 LPC cepstrum coefficients (i.e. 12 variables). The task is to distinguish nine male speakers by their utterances. Speakers 1 to 9 have the number of observations: 61, 65, 118, 74, 59, 54, 70, 80, 59, respectively. The time length we use is 7 and the data dimension is 7×12.
- WAFER dataset (WAF). WAF has 327 observations of two classes. Each observation is labeled as normal or abnormal and recorded by six vacuum-chamber sensors (i.e. variables) during monitoring an operational semiconductor fabrication plant. The normal and abnormal classes have 200 and 127 observations, respectively. The time length we use is 104 and thus the observation size is 104×6.
- BCI dataset. BCI contains a total of 416 observations, which has been further divided into a training set of 316 observations and a test set of 100 observations. For convenience, let $B_{tr} = 316/416$. BCI consists of two classes: upcoming left and right hand movements, each 208 observations. Each observation is collected using 28 EEG channels (i.e. 28 variables). The time length is 500 and the dimension is 500×28.

Table 2 summarizes several statistics of the five datasets used in our experiments.

Table 3
The lowest average error rates shown as (mean±std(dim.)) and their corresponding dimensionalities by different methods on the five MTS datasets. The best method is shown in bold face. • means that RBLDA is significantly better than the method, and ∘ means that the difference between RBLDA and the method is not significant, using the Wilcoxon signed-rank test with a p-value of 0.05.

Data	p	Method					
		BPCA	BLDA	PBLDA	RLDA	RBLDA	DTW
AUS	p_1	15.5±2.0(1,21)•	5.2±2.5(1,16)•	5.2±2.5(1,16)•	7.7±2.4(14)•	**2.9**±0.7(1,13)	26.2±2.2•
	p_2	11.0±1.4(1,21)•	1.9±0.5(1,15)∘	1.9±0.5(1,15)∘	2.3±1.1(16)∘	**1.7**±0.6(1,14)	19.4±1.7•
ECG	p_1	25.5±4.8(15,1)∘	—-	36.6±7.9(1,2)•	23.6±6.5(1)∘	**23.6**±5.8(2,2)	27.0±4.0•
	p_2	**13.4**±5.4(8,1)∘	16.3±7.0(2,2)∘	16.3±7.0(2,2)∘	20.7±3.8(1)•	14.9±5.3(2,1)	19.0±5.1•
JAP	p_1	22.4±3.4(2,12)•	20.2±5.5(2,8)∘	20.2±5.5(2,8)∘	18.5±5.7(8)∘	**17.4**±4.1(1,12)	20.5±2.3•
	p_2	7.2±2.8(3,11)•	4.5±1.5(2,9)∘	4.5±1.5(2,9)∘	4.9±2.2(8)∘	**3.9**±1.4(3,12)	9.0±2.6•
WAF	p_1	21.2±2.9(9,6)•	7.1±2.0(2,5)∘	7.1±2.0(2,5)∘	12.7±2.3(1)•	7.7±2.6(2,3)	**7.0**±1.1∘
	p_2	11.1±3.6(12,6)•	**2.9**±1.6(4,5)∘	**2.9**±1.6(4,5)∘	9.6±4.3(1)•	3.0±2.0(6,4)	3.9±2.6∘
BCI	p_1	48.3±2.4(1,8)•	—-	45.1±3.6(24,7)•	49.1±5.4(1)•	**42.5**±6.5(3,2)	48.8±3.8•
	p_2	42.0±0.0(4,5)•	34.0±0.0(16,25)•	34.0±0.0(16,25)•	32.0±2.5(1)•	**25.7**±2.8(8,1)	47.0±0.0•

4.1. Classification performance on real MTS datasets

In this subsection, we compare the classification performance of RBLDA and related competitors including BLDA [4], PBLDA [16], RLDA [20], BPCA [3] and DTW [13]. For BLDA, we use the implementation in [21] as detailed in Sec. 2.6.1. For RBLDA, we use Algorithm 2 and choose the two regularization parameters r_1, r_2 from the set $\{10^{-6}, 0.001, 0.01, 0.1, 0.2, \ldots, 0.9, 0.99\}$. For RLDA, we use a model selection algorithm similar to Algorithm 2 and choose the regularization parameter r from the same set as RBLDA. Let $\mathbf{V}_w = (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_q)$ be the discriminant transformation obtained by RLDA with Algorithm 1. We also examine the performance of using $\mathbf{V}_w^u = (\mathbf{v}_1^u, \mathbf{v}_2^u, \ldots, \mathbf{v}_q^u)$, where $\mathbf{v}_i^u = \mathbf{v}_i/||\mathbf{v}_i||_2$ (i.e., the length of $\mathbf{v}_i^u$ equals to 1). For RLDA, we report the better result between $\mathbf{V}_w$ and $\mathbf{V}_w^u$. For BLDA, PBLDA and RBLDA, we also report similar results.

To measure the misclassification rate, each dataset is randomly split into a training set containing a ratio of $p_i, i = 1, 2$ of all samples per class and a test set containing the remaining samples. Several values of p_i for each data set are investigated, which is listed in Table 2. Particularly, as mentioned in the Sec. 1, the small-sample-size training is an important factor in comparing matrix-based and vector-based methods, a small p_1 and a large p_2 are hence respectively used to investigate the performance in small and large training sample size cases. Furthermore, due to the varying levels of difficulty in classification among the five datasets, similar to that in [18], we use irregular values for p_1 and p_2 so that the error rates on different datasets could be roughly comparable.

For all methods, the 1-nearest-neighbor classifier is run in the reduced-dimensional space to obtain the test misclassification rates. All possible dimensionalities of the reduced representation are tried and the lowest average misclassification rate from 10 random splittings, its corresponding latent dimension, and standard deviation are reported. The regularization parameters involved in RLDA and RBLDA on each splitting are learned by 5-fold cross validation. For BCI, we always use the standard test set consisting of 100 observations to compute test misclassification rates.

The classification results on the five MTS datasets are shown in Table 3, where the '—-' sign means that the method fails to run, and hence the misclassification rate is not available. The detailed results with different dimensionalities in low-dimensional space are visualized in Fig. 1. For RLDA, we plot the misclassification rates versus the dimensionality of $\mathbf{y} = \mathbf{V}_w'\mathbf{x}$. For the $\mathbf{Y} = \mathbf{V}_1'\mathbf{X}\mathbf{V}_2$ in bilinear methods, we plot the misclassification rates versus the dimensionality of each $\mathbf{y}_j$, where $\mathbf{y}_j$ is the j-th column of $\mathbf{Y} = (\mathbf{y}_1, \ldots, \mathbf{y}_{d_2})$. Since the size of $\mathbf{Y}$ on AUS or BCI is a little large, only an upper-left submatrix of $\mathbf{Y}$ that includes the optimal result is shown: 15×22 for AUS, 24×10 for BCI. For better comparison, we also add the result of DTW in Fig. 1 although it actually uses all the original data dimensions.

Denote the difference in misclassification rates between RBLDA and a *method* by $\delta = e^{RPBLDA} - e^{method}$. To examine whether the difference δ is statistically significant, we perform the Wilcoxon signed-rank test at two levels: *Test1* and *Test2*. The null and alternative hypotheses are $H_0 : \delta = 0$ vs $H_1 : \delta < 0$.[3] *Test1* is used to test the significance at the level of each individual (p_i, $dataset_j$), $i = 1, 2, j = 1, 2, \ldots, 5$, using the 10 misclassification rates of a *method* corresponding to the optimal dimension from 10 random splittings, i.e., $e_l^{method}, l = 1, \ldots, 10$. The results for *Test1* are also given in Table 3. *Test2* is used to test the overall significance at the level of all ten individuals. This is performed by using the 100 misclassification rates of a *method* collected from all individuals ($p_i, dataset_j$), i.e., $e_l^{method}, i = l, \ldots, 100$. For BLDA, only 80 misclassification rates are available for *Test2*. The results for *Test2* are summarized in Table 4. From Table 4, it can be seen that RBLDA is the overall winner of 10 competitions.

For individual comparison, the main observations from Table 3 and Fig. 1 include

(i) RBLDA vs. RLDA. RBLDA performs better than RLDA, and 6 out of 10 competitions are significant, which means that the utilization of matrix data structure is useful for MTS data classification.

[3] If the test is not rejected, a second test $H_0 : \delta = 0$ vs $H_1 : \delta > 0$ will be performed. If the second test is not yet rejected, it is concluded that the difference between two methods are not significant.

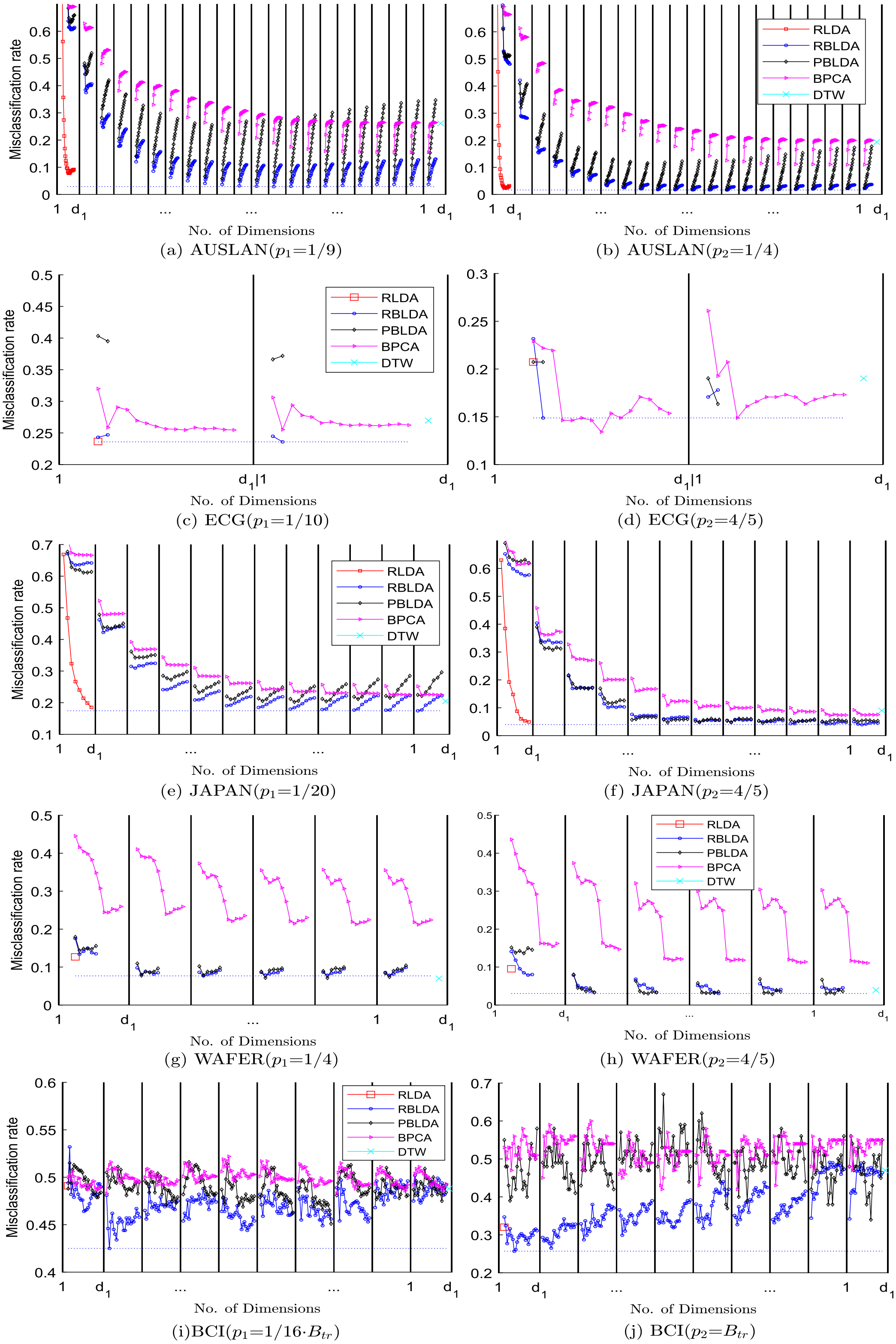

Fig. 1. Misclassification rates versus number of dimensions by different methods in 10 competitions on the five datasets. Note that the marker x signals the result of DTW that actually uses all the original data dimensions.

Table 4
Results of p-values between RBLDA and other methods by the Wilcoxon signed-rank test. The superiority of RBLDA over a method is highly significant if p-value< 0.001.

Method				
BPCA	BLDA	PBLDA	RLDA	DTW
3.1e-12	3.6e-04	1.1e-5	3.0e-9	4.5e-13

(ii) RBLDA vs. BLDA and PBLDA. RBLDA is comparable with or superior to BLDA and PBLDA, and 4 out of 10 competitions are significant. RBLDA is more advantageous in small sample size cases on AUS, ECG, and BCI, where RBLDA has significantly better performance than PBLDA while BLDA even fails to run on ECG and BCI. Furthermore, it can be observed from Fig. 1 that, in most cases, RBLDA outperforms PBLDA not only in the optimal dimension but also in a wide range of dimensions. This indicates that the gain of applying regularization to BLDA is substantial.

(iii) RBLDA vs. BPCA. RBLDA is substantially better than BPCA except for p_2 on ECG data, which means that the utilization of label information is beneficial to MTS data classification in general.

(iv) RBLDA vs. DTW. RBLDA significantly outperforms DTW on all five datasets except for WAFER. Their performance on WAFER is roughly comparable.

4.2. Low-dimensional plots via RLDA and RBLDA

In this subsection, we compare the performance of RLDA and RBLDA for data visualization. To this end, we choose the two-class datasets ECG ($p = 4/5$) and WAFER ($p = 4/5$). The popularity of RLDA is partly due to the reduced-rank constraint that enables us to view informative low-dimensional projections of the data [25]. Since $\text{rank}(\mathbf{S}_b) \le c-1$, the dimensionality of RLDA subspace is at most $c-1$. If d is much larger than c, the dimension reduction will be substantial. In particular, in the case that $c = 2$, the dimensionality of the subspace is only 1. That is, ECG and WAFER can be viewed in and only in a one-dimensional RLDA subspace.

However, this is not the case for RBLDA. Since $\text{rank}(\mathbf{S}_{1b}) \le \min(d_1, d_2(c-1))$, $\text{rank}(\mathbf{S}_{2b}) \le \min\ (d_1(c-1), d_2)$, the dimensions of RBLDA subspace are at most 2×2 on ECG and 6×6 on WAFER, respectively. This means that more discriminant features are available with RBLDA than those with RLDA. It is thus interesting to investigate whether or not the more discriminant features are useful for data visualization. For demonstration purposes, we use one result from 10 random spitting which is similar to the average one in Table 4.

In Fig. 2, we plot the one-dimensional and two-dimensional projections of the remaining 1/5 test data obtained by RLDA and RBLDA, respectively. In Fig. 2 (b) and (d), we choose only two discriminant features y_{21} and y_{22} from the 2×2 and 6×6 projected $\mathbf{Y}$, respectively. It can be seen from Fig. 2 that RBLDA yields better class separation than RLDA and more discriminant features extracted by RBLDA are beneficial to data visualization. From Fig. 1 (d) and (h), it can be observed that the more discriminant features available in RBLDA can improve the classification.

4.3. Performance on computational efficiency

In this subsection, we examine the computational efficiency of Implementation 2 in Sec. 3.3.1 and Algorithm 2 using BCI data, for which $d_1 = 500$ and $d_2 = 28$.

4.3.1. Efficient implementation for a fixed value of (r_1, r_2)

To address the small training sample size case, a subset is randomly chosen from BCI with a sample size $n_1 = 18$, forming a new dataset called BCI1. To investigate the performance when d_1 is much greater than d_2, b datasets are constructed from BCI1, each of which is created by replicating the data along the row direction b times, resulting in BCI2-b with $d_1 = 500b$. Next, to investigate the performance with a larger sample size, the standard training set of BCI is used and the observations are repeated 20 times and the data is then replicated along the row direction 8 times, yielding a new dataset called BCI3, of size $4000 \times 28 \times 6320$.

The computation times on BCI1, BCI2-8 and BCI3 among five closely related methods or implementations are summarized in Table 5, including RLDA by direct implementation (6), RLDA by Algorithm 1, PBLDA [16], one/two-parameter RBLDA by direct implementation (18), and two-parameter RBLDA by Implementation 2. Furthermore, to examine the performance when d_1 is much larger than d_2, Fig. 3 plots the computational times on BCI2-b by four of these methods versus various values of b.

It can be observed from Table 5 and Fig. 3 that

(i) efficient implementation vs. direct implementation: In $d \gg n$ scenarios, Algorithm 1 of RLDA is computationally more efficient than the direct implementation. Similarly, when d_1 is much larger than d_2, the proposed Implementation 2 of RBLDA exhibits considerably improved computational efficiency compared to the direct implementation (18), especially when $b \ge 4$ on BCI2-b.

(ii) RBLDA vs. RLDA: on BCI3, where the sample size becomes large, the proposed efficient implementation of RBLDA (i.e., Implementation 2) can be computationally more efficient that of RLDA (i.e., Algorithm 1).

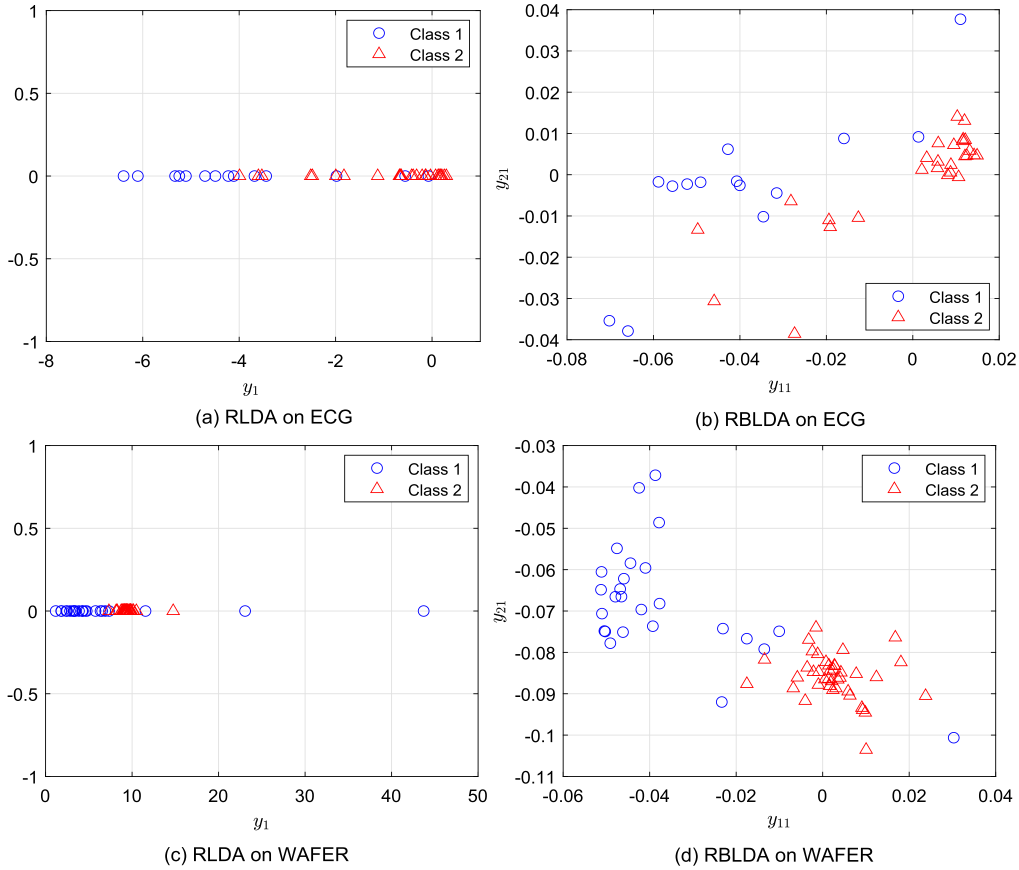

Fig. 2. The one-dimensional, two-dimensional plots by RLDA, RBLDA on ECG and WAFER datasets, respectively.

Table 5
Training time for a fixed regularization parameter by different methods on the three constructed datasets.

method	dataset ($d_1 \times d_2 \times n$)		
	BCI1 $500 \times 28 \times 18$	BCI2 $4000 \times 28 \times 18$	BCI3 $4000 \times 28 \times 6320$
RLDA: direct	1429.6	—-	—-
RLDA: efficient [19,20]	0.004	0.03	102.17
PBLDA [16]	0.031	0.12	33.61
one/two-parameter RBLDA: direct	0.055	8.59	35.33
two-parameter RBLDA: Impl. 2	0.027	0.11	33.06

Table 6
Comparison of time used by RBLDA model selection algorithm on BCI data with different dimensions.

Dimension	Number of candidates m^2						$T(100,100)/$ $T(1,1)$	Traditional CV $(100,100)/$ $T(100,100)$
	1	4	25	100	2500	10000		
500×28	0.11	0.13	0.15	0.31	2.71	7.61	69.2	144.5
4000×28	0.37	0.38	0.52	0.60	3.27	9.96	26.9	371.5
16000×28	2.19	2.14	2.42	2.67	6.85	21.25	9.7	1030.6

4.3.2. RBLDA model selection algorithm

As mentioned in Sec. 3.3.2, the traditional CV is expensive to use for RBLDA. The objective of this experiment is to investigate the efficiency of the proposed RBLDA model selection algorithm, i.e., Algorithm 2, compared to traditional CV. For simplicity, we use the same number of candidates for r_1 and r_2, that is, $m = m_1 = m_2$. Various values of m are tried and the computational times on BCI1, BCI2-8 and BCI2-32 are recorded in Table 6.

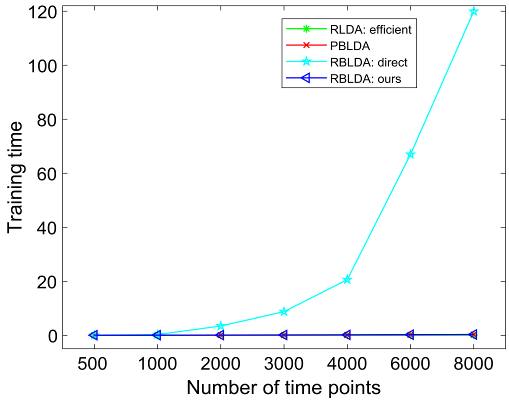

Fig. 3. Training time versus number of time points by different methods.

It can be seen from Table 6 that

(i) as expected, the proposed RBLDA model selection algorithm is much more efficient than the traditional CV procedure. Although the candidate size increases by 10000 fold, the increases in running time are only 69.2 on BCI and 9.7 on BCI3, rather than 10000.

(ii) the proposed Algorithm 2 is more efficient when d_1 is much greater than d_2.

5. Conclusions and discussions

In this paper, we investigate the critical question of whether the superiority of vector-based RLDA over matrix-based BLDA would always hold for general matrix data by means of breaking two previous limitations. For the first limitation of data type, we focus on another kind of matrix data, namely MTS data. For the second limitation of method type, we propose a new two-parameter RBLDA and develop an efficient model selection algorithm, which enables us to compare the two kinds of methods fairly. We also explore how to perform visualization on MTS data using RBLDA.

In contrast to the existing results on image data [6,7] and the new results in Appendix B, our empirical evidence on MTS data shows that the superiority of vector-based RLDA over matrix-based BLDA does not always hold for general matrix data. Instead, we have found that the proposed RBLDA generally performs better than RLDA on MTS data. Therefore, besides the known factor of small-sample-size training, a more critical factor to the superiority is whether the assumption of a separable transformation matrix could be supported by the data under study. From our results, it could be concluded that the separability assumption is more suitable for MTS data than that for image data. We believe that our findings in this paper may inspire new developments and more applications of matrix-based methods to MTS data. For future work, it is meaningful to develop some measures or statistical tests to assess separability and evaluate its impact on discriminant capability.

A distinct feature of MTS data from image data is that the observations commonly have the same number of variables but may have different numbers of time points. Namely, the time length could be different. It would be interesting to extend our proposed RBLDA to accommodate such data, which is one of our future works.

In recent years, robust extensions of 2DPCA and 2DLDA using various techniques have received much attention, including robust 2DLDA and BLDA using L_p-norm [26,27], robust 2DLDA via Information Divergence [28], 2DPCA using the nuclear-norm [29], robust tensor LDA based on the Laplace distribution [30], 2DLDA using the nuclear-norm [31], trace ratio 2DLDA using l_1-norm [32], and etc. However, most of these methods are limited to image data. It would be worthwhile to extend these methods to MTS data classification in the future.

CRediT authorship contribution statement

Jianhua Zhao: Conceptualization, Funding acquisition, Methodology, Supervision, Writing – review & editing. **Haiye Liang:** Formal analysis, Investigation, Methodology, Validation, Writing – original draft. **Shulan Li:** Investigation, Methodology, Validation, Writing – original draft. **Zhiji Yang:** Funding acquisition, Methodology, Writing – review & editing. **Zhen Wang:** Funding acquisition, Resources, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We have shared the link to our data at the footnote of page 9 in this paper.

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Appendix A. The computation in row direction

We denote

$$\begin{aligned}
\mathbf{X}_{(2)} &= [\mathbf{X}_1', \dots, \mathbf{X}_n'] \ (d_2 \times d_1 n),\\
\mathbf{W}_{(2)} &= [\mathbf{W}_1', \dots, \mathbf{W}_c'] \ (d_2 \times d_1 c),\\
\mathbf{\Pi}_2 &= \mathrm{blkdiag}(n_1\mathbf{I}_{d_1}, \dots, n_c\mathbf{I}_{d_1}) \ (d_1 c \times d_1 c),\\
\mathbf{\Pi}_2^{1/2} &= \mathrm{blkdiag}(\sqrt{n_1}\mathbf{I}_{d_1}, \dots, \sqrt{n_c}\mathbf{I}_{d_1}) \ (d_1 c \times d_1 c),\\
\boldsymbol{\pi}_2 &= (n_1\mathbf{I}_{d_1}, \dots, n_c\mathbf{I}_{d_1})' \ (d_1 c \times d_1),\\
\sqrt{\boldsymbol{\pi}_2} &= (\sqrt{n_1}\mathbf{I}_{d_1}, \dots, \sqrt{n_c}\mathbf{I}_{d_1})' \ (d_1 c \times d_1),\\
\mathbb{1}_{2m} &= (\mathbf{I}_{d_1}, \dots, \mathbf{I}_{d_1})' \ (d_1 m \times d_1),\\
\mathbf{H}_2 &= \mathbf{I}_{d_1 n} - \frac{1}{n}\mathbb{1}_{2n}\mathbb{1}_{2n}' \ (d_1 n \times d_1 n),\\
\mathbf{H}_{2\pi} &= \mathbf{I}_{d_1 c} - \frac{1}{n}\sqrt{\boldsymbol{\pi}_2}\sqrt{\boldsymbol{\pi}_2}' \ (d_1 c \times d_1 c),
\end{aligned}$$

$\mathbb{E}_2 = (\mathbf{E}_{ij})$ $(d_1 n \times d_1 c)$ be a $n \times c$ partitioned matrix with the $d_1 \times d_1$ ij-th submatrices $\mathbf{E}_{ij} = \mathbf{I}_{d_1}$ if $\mathbf{X}_i$ belonging to class j and $\mathbf{E}_{ij} = \mathbf{0}_{d_1}$ otherwise.

Assuming that $\mathbf{X}_{(2)} = \mathbf{X}_{(2)}\mathbf{H}_2$, $\mathbf{S}_{2b}$ in (16) is rewritten as

$$\mathbf{S}_{2b} = \mathbf{X}_{(2)}\mathbb{E}_2\mathbf{\Pi}_2^{-1}\mathbb{E}_2'\mathbf{X}_{(2)}'. \tag{A.1}$$

The condensed SVD of $\mathbf{S}_{2t}$ in (20) is

$$\mathbf{S}_{2t} = \mathbf{X}_{(2)}\mathbf{X}_{(2)}' = \mathbf{U}_{2t}\mathbf{\Gamma}_{2t}\mathbf{U}_{2t}', \tag{A.2}$$

where $t_2 = \mathrm{rank}(\mathbf{\Gamma}_{2t})$. Accordingly, the EVD of $\mathbf{S}_{2t}^{r_2}$ in (22) is given by

$$\mathbf{S}_{2t}^{r_2} = \mathbf{U}_{2t}\mathbf{\Gamma}_{2t}^{r_2}\mathbf{U}_{2t}' + r_2\hat{\sigma}_2^2\mathbf{U}_{2t}^{\perp}\mathbf{U}_{2t}^{\perp\prime}, \tag{A.3}$$

where $\mathbf{\Gamma}_{2t}^{r_2} = (1 - r_2)\mathbf{\Gamma}_{2t} + r_2\hat{\sigma}_2^2\mathbf{I}_{d_1}$, and $\mathbf{U}_{2t}^{\perp}$ is the orthogonal complement of $\mathbf{U}_{2t}$. Let

$$\mathbf{X}_{(2u)} = \mathbf{U}_{2t}'\mathbf{X}_{(2)}. \ (t_2 \times d_1 n) \tag{A.4}$$

By (A.3), the regularized total scatter matrix in $\mathbf{X}_{(2u)}$-space is

$$\mathbf{S}_{2tu}^{r_1} = \mathbf{\Gamma}_{2t}^{r_2}.$$

Define

$$\mathbf{F}_{2bu} = \mathbf{X}_{(2u)}\mathbb{E}_2\mathbf{\Pi}_2^{-1/2}. \ (t_2 \times d_1 c) \tag{A.5}$$

(i) When $d_1 c \geq d_2$, let

$$\mathbf{R}_{21} = \mathbf{\Gamma}_{2t}^{r_2\,-1/2}\mathbf{F}_{2bu}\mathbf{F}_{2bu}'\mathbf{\Gamma}_{2t}^{r_2\,-1/2}. \tag{A.6}$$

If $(\mathbf{\Lambda}_2, \mathbf{V}_{2R})$ is the eigenpair of $\mathbf{R}_{21}$ and $\mathbf{V}_{2u} = \mathbf{\Gamma}_{2t}^{r_2\,-1/2}\mathbf{V}_{2R}$, then $(\mathbf{\Lambda}_2, \mathbf{V}_{2u})$ is the solution in $\mathbf{X}_{(2u)}$-space.
(ii) When $d_1 c < d_2$, let

$$\mathbf{G}_{22u} = \mathbf{\Gamma}_{2t}^{r_2\,-1}\mathbf{F}_{2bu}, \tag{A.7}$$

$$\mathbf{R}_{22} = \mathbf{F}_{2bu}'\mathbf{\Gamma}_{2t}^{r_2\,-1}\mathbf{F}_{2bu}. \tag{A.8}$$

If $(\mathbf{\Lambda}_2, \mathbf{V}_{2R})$ is the eigenpair of $\mathbf{R}_{22}$, and $\mathbf{V}_{2u} = \mathbf{G}_{22u}\mathbf{V}_{2R}$ then $(\mathbf{\Lambda}_2, \mathbf{V}_{2u})$ is solution in $\mathbf{X}_{(2u)}$-space.

Table B.7
The lowest average error rates shown as (mean±std(dim.)) and their corresponding dimensionalities by different methods on the three face datasets. γ: No. of training images per individual. The best method is shown in bold face. * means that RLDA is significantly better than the method, and ⋄ means that the difference between RLDA and the method is not significant, using the Wilcoxon signed-rank test with a p-value of 0.05.

Data	γ	Method		
		RLDA	RBLDA	BLDA
YALE	3	**16.5**±4.4(14)	21.5±5.1(18,8)*	35.0±7.0(7,11)*
	5	**12.5**±5.5(14)	16.6±6.6(9,31)*	22.2±6.0(8,10)*
	7	**9.7**±5.7(14)	11.8±7.0(17,16)*	15.3±7.9(10,11)*
CMU PIE	3	**9.4**±8.3(67)	10.7±10.9(18,9)⋄	10.9±10.8(33,8)*
	4	**3.0**±3.5(67)	4.2±5.0(25,13)*	4.4±4.5(30,7)*
	5	**2.4**±2.6(67)	3.4±3.1(27,14)*	3.9±4.2(28,8)*
XM2VTS	3	**4.6**±1.6(294)	7.0±1.9(15,15)*	6.9±1.9(18,15)*
	4	**2.6**±1.5(67)	4.9±1.7(18,12)*	4.8±1.7(17,10)*
	5	**1.7**±1.0(96)	3.7±1.5(14,15)*	3.6±1.4(17,13)*

Table B.8
Results of p-values between RLDA and other methods by the Wilcoxon signed-rank test on the three face datasets. The superiority of RLDA over a method is highly significant if p-value< 0.001.

Method	
RBLDA	BLDA
1.88e-39	3.18e-56

Appendix B. Classification performance on face datasets

It has been reported in [6,7] that RLDA outperforms the matrix-based BLDA on a number of real image datasets. In this experiment, we futher use image data to compare the classification performance of RLDA and the proposed two-parameter RBLDA. Unlike the existing results that reported the performance of RLDA with a fixed value of the regularization parameter [6,7], our comparison adopts the same experimental procedure used for the MTS data as detailed in 4.1. That is, we employ cross-validation to select the regularization parameter involved in RLDA and RBLDA. For comparison, the performance BLDA is also added.

We use the following three face data sets.

- Yale database[4] comprises 165 images of 15 individuals, with each person having 11 images. These images capture various facial expressions or configurations, such as center-light, with glasses, happy, left-light, without glasses, normal, right-light, sad, sleepy, surprised, and wink. We subsample the images to the size 64×64.
- CMU PIE[5] face database is comprised of 41,368 face images featuring 68 subjects. Each subject's facial images were captured in 13 different poses, 43 distinct illumination conditions, and with 4 varied expressions. For our experiments, we utilize the same subdatabase as described in [7]. This subdatabase consists of 43 images per person, specifically capturing frontal face images under different lighting conditions while excluding expression variations. We subsample the images to the size 64×64.
- XM2VTS database[6] consists of 2360 images featuring 295 individuals. Each individual is represented by 8 images that were captured over a span of four months. The image size is 51×55.

Each dataset is divided into a training set and a test set. To accomplish this, a random selection of γ images is made from each person, forming the training set. The remaining images are then allocated for use as the test set. For each data set, three values of γ chosen from the set $\{3, 4, 5, 7\}$ are investigated. The results from 50 random splittings are summarized in Table B.7 and Table B.8.

The following can be observed from Table B.7 and Table B.8.

(i) From Table B.8, RLDA is the overall winner on the three image datasets.
(ii) From Table B.7, RLDA performs significantly better than RBLDA in each of the 15 competitions, except for the case $\gamma = 3$ on CMU PIE, where the test result is not significant. This observation is in strong contrast to the findings observed on MTS datasets in 4.1.

[4] Available from http://cvc.yale.edu/projects/yalefaces/yalefaces.html.
[5] Available from http://www.ri.cmu.edu/projects/project_418.html.
[6] Available from http://www.face-rec.org/databases/.

(iii) RBLDA demonstrates significant performance superiority over BLDA on YALE, while both methods exhibit similar performance on the other two datasets.

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